

Mohamed Shehab-Eldin Mohamed

Mechatronic Engineering Student | Robotics, Automation & Control Systems

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Professional Summary

Final-year Mechatronic Engineering student at Asia Pacific University with practical experience in robotics, control systems and industrial automation, gained through a 16-week placement in steel fabrication and academic projects including a ROS 2 autonomous service robot. Seeking an engineering internship where strong technical fundamentals and a disciplined approach to problem-solving can contribute to real engineering work.

Education

Bachelor of Engineering in Mechatronics

- Asia Pacific University (APU) – Malaysia
- Study Period : 2022 – 2027
- CGPA : 3.35

Professional Experience

Production Engineering Intern — Ferrometalco Steel Construction (FMC)

Cairo, Egypt | 16-week industrial placement, Sep 2025 – Jan 2026

- Worked in the fabrication workshop on tower crane mast sections, following the full production cycle from material receiving through cutting, drilling, fit-up, welding, painting and shipment.
- Interpreted shop drawings and carried out dimensional inspections with the QC team, checking lengths, squareness and bolt hole positions against drawing tolerances.
- Monitored SMAW and MIG welding and distortion control, and followed non-conformance reports and the corrective actions taken on the shop floor.
- Awarded a Certificate of Achievement by the company for excellent work during the placement.

Engineering Projects

Autonomous Indoor Delivery Service Robot Using ROS 2 for Structured University Environments

Final Year Project — APU Centre of Robotics Excellence (APCORE)

- Designed and developed an autonomous service robot that navigates a university building on its own to deliver items and guide visitors.
- Built the voice-based chatbot interface used to request deliveries and monitor the robot, and evaluated the complete system through repeated live trials.

Arduino-Based Robot Car

- Designed and programmed a robot car that follows a path using infrared and ultrasonic sensors for obstacle detection and path tracking.

PLC-Based Cork Cutter System

- Developed a simulated cork cutting system controlled by PLC logic to demonstrate automation principles and sequencing operations.

Tools & Software

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|------------------|---------------------|
| • ROS 2 (Humble) | • SolidWorks |
| • Python | • MATLAB & Simulink |
| • C Programming | • Linux / Ubuntu |
| • Arduino IDE | • Git & GitHub |

Skills

Technical Skills

- Autonomous mobile robotics and ROS 2 development
- Embedded programming, sensors and actuators
- PLC fundamentals, control systems, circuit design
- Quality control, inspection and drawing interpretation
- Languages: Arabic (native), English (very good)

Professional Skills

- High adaptability and willingness to learn
- Good time management and organization
- Strong team collaboration and communication
- Problem-solving mindset with attention to detail
- Curious about modern engineering trends